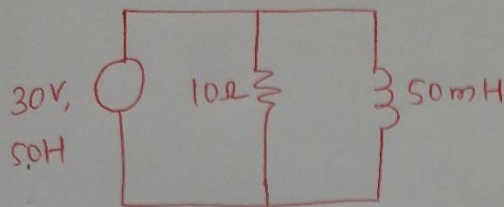


* For the following parallel R-L ckt, find the phase diff between V & I . Also find active power & the power factor of the ckt.



$$X_L = j\omega L$$

$$= j2\pi f \times 50\text{mH}$$

$$= j2\pi \times 50 \times 50\text{mH}$$

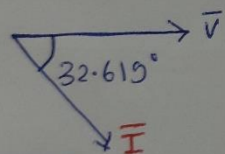
$$|X_L| = 15.707\Omega$$

$$\bar{Y} = \frac{1}{R} + \frac{1}{j\omega L}$$

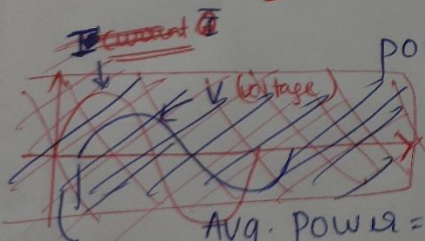
$$\bar{Y} = \frac{1}{10} - \frac{j}{15.707} = \underline{\underline{0.1 - 0.064j}}$$

$$\bar{I} = \frac{V}{Z} = V\bar{Y} = 50 [0.1 - 0.064j]$$

$$\bar{I} = 5 - 3.2j = 5.936 \angle -32.619^\circ \text{ A}$$



In circuit, current lags the voltage by 32.619° .



$$\phi = 32.615^\circ$$

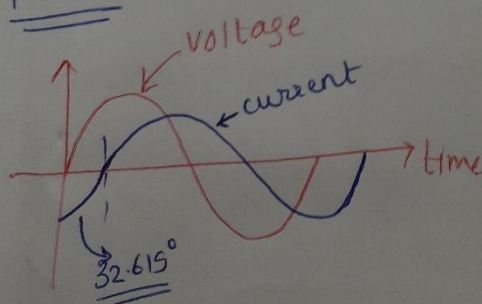
$$\text{power factor} = \cos \phi$$

$$= \cos(32.615) = 0.8422 \text{ (Lagging)}$$

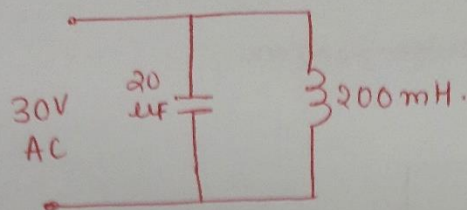
$$\text{Avg. Power} = \text{active power} = V_{\text{rms}} I_{\text{rms}} \cos \phi$$

$$= 30 \times 5.936 \times 0.8422$$

$$= \underline{\underline{149.97 \text{ W}}}$$



- For the following parallel L-C ckt, find the resonating frequency & the current flowing through the inductor

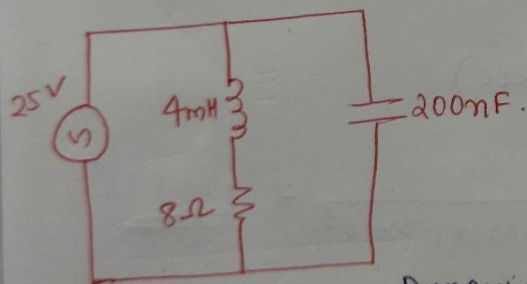


Solⁿ:-
$$f_r = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{200 \times 10^{-3} \times 20 \times 10^{-6}}} = 79.578 \text{ Hz}$$

$$|I_L| = \frac{V_L}{|X_L|} = \frac{30}{|j2\pi \times 79.578 \times 200 \times 10^{-3}|}$$

$$|I_L| = \frac{30}{100} = \underline{\underline{0.3 \text{ A}}}$$

- For the following ckt, find resonant frequency, dynamic impedance, total current at resonant & quality factor, Q.



Solⁿ:-
$$f_r = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

$$= \frac{1}{2\pi} \sqrt{\frac{1}{4 \times 200 \times 10^{-12}} - \frac{8^2}{16 \times 10^{-6}}}$$

$$\boxed{f_r = 5.618 \text{ kHz}}$$

Dynamic impedance $Z_0 = \frac{L}{CR} = \frac{4 \times 10^{-3}}{8 \times 200 \times 10^{-9}} =$

$$\boxed{Z_0 = 2.5 \text{ k}\Omega}$$

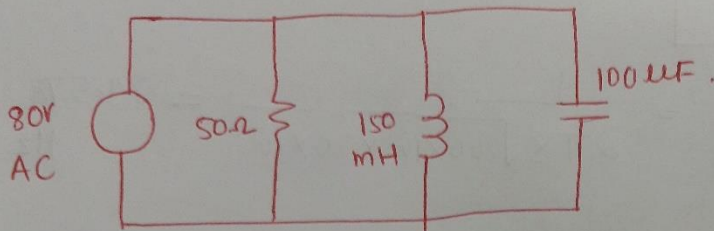
Current at resonant condⁿ = $\frac{25}{Z_0} = \frac{25}{2.5 \text{ k}} = \underline{\underline{10 \text{ mA}}}$

Quality factor = $\frac{\omega_r \cdot L}{R} = \frac{2\pi f_r L}{R} = \frac{2\pi \times 5.618 \times 10^3 \times 4 \times 10^{-3}}{8}$

$$\boxed{Q = 17.649} \checkmark$$

For the following parallel R-L-C circuit

- find
- i) Resonant frequency
 - ii) Bandwidth
 - iii) Quality factor
 - iv) Current at resonant
 - v) ~~At~~ Current magnification.



Sol:- $f_{or} = \frac{1}{2\pi\sqrt{LC}} = \underline{41.093 \text{ Hz}}$

$Q = \left[\frac{\omega_r L}{R} \right] = \frac{2\pi \times 41.093 \times 150 \times 10^{-3}}{50} = \underline{0.77}$ ~~(poor Q)~~

~~Bandwidth = $\frac{f_r}{Q}$~~ Ignore

$$Q = \frac{R}{X_L} \quad (\text{for parallel RLC})$$

$$= \frac{50}{2\pi f_r X_L} = \frac{50}{2\pi \times 41.093 \times 150 \times 10^{-3}} = \underline{1.291}$$

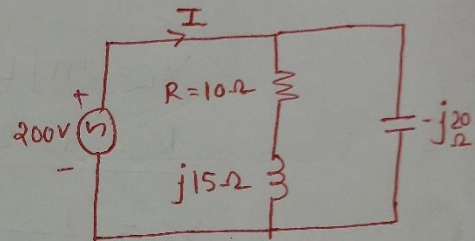
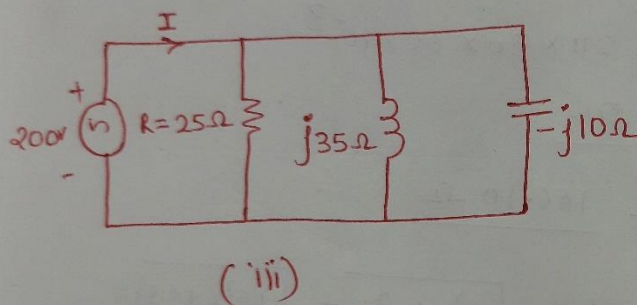
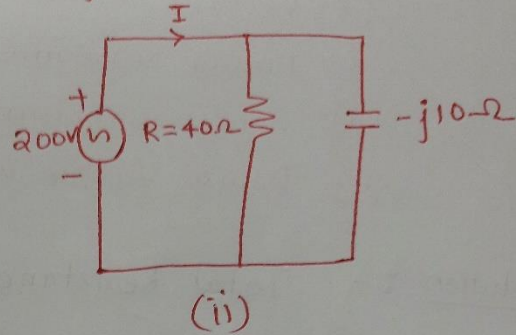
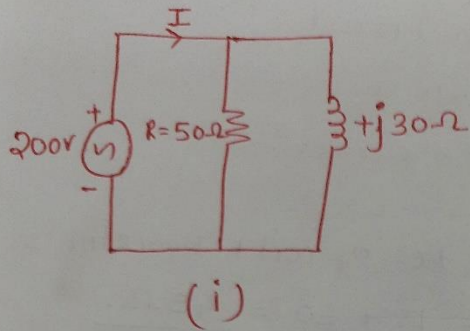
$$B.W = \frac{f_r}{Q} = \underline{31.83 \text{ Hz}}$$

$$\text{Current at resonant} = \frac{V}{R} = \frac{80}{50} = 1.6 \text{ A.}$$

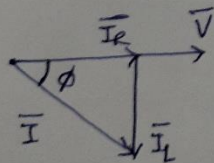
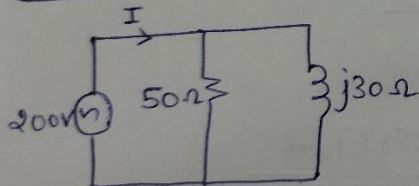
$$\begin{aligned} \text{Current magnification} &= Q \times I = 1.291 \times 1.6 = \\ &= \underline{2.0656 \text{ A.}} \end{aligned}$$

* Parallel AC circuit :-

* For the following circuits, $V = 200\text{V AC}$.
Find the current, I , power factor.



Solution :- for (i)



$$Y = \frac{1}{50} + \frac{1}{j30} = 0.02 - j0.033$$

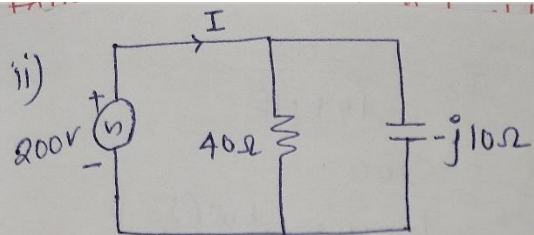
$$\bar{I} = V \bar{Y}$$

$$\bar{I} = 200 [0.02 - j0.033]$$

$$\bar{I} = 4 - 6.6j \text{ A}$$

$$\phi = \tan^{-1} \left(\frac{6.6}{4} \right) = 58.78^\circ$$

$$\text{power factor } \cos \phi = 0.518 \text{ (lagging).}$$



$$\bar{Y} = \frac{1}{40} + \frac{j}{10\Omega}$$

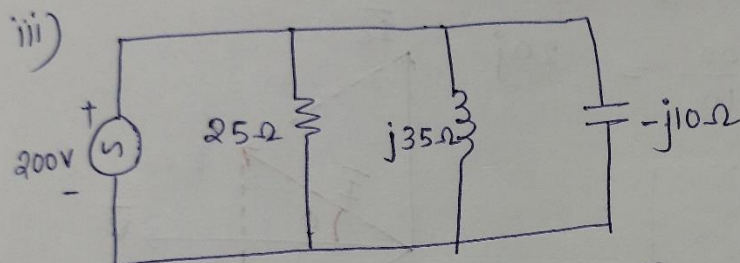
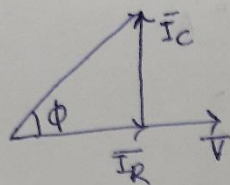
$$\bar{Y} = 0.025 + j0.1$$

$$\bar{I} = V\bar{Y} = 200[0.025 + j0.1]$$

$$\bar{I} = 5 + j20 \text{ A}$$

$$\phi = \tan^{-1}\left(\frac{20}{5}\right) = 75.96^\circ$$

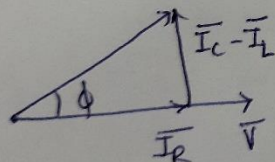
power factor = $\cos \phi = \underline{0.2425}$



$$\bar{Y} = \frac{1}{25} + j\left[\frac{1}{10} - \frac{1}{35}\right] = 0.04 + j0.0714$$

$$\bar{I} = V\bar{Y} = 200[0.04 + j0.0714]$$

$$\bar{I} = 8 + 14.28j \text{ A}$$



$$\phi = \tan^{-1}\left[\frac{14.28}{8}\right] = 60.741^\circ$$

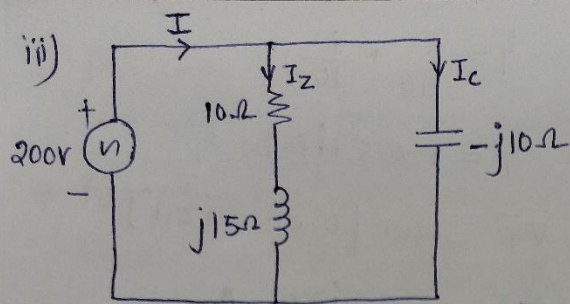
power factor = $\cos \phi = \underline{0.488 \text{ (leading)}}$

ii) In all three cases, I_R , I_L & I_C can be easily found using $I_R = \frac{V}{R}$, $I_L = \frac{V}{jX_L}$, $I_C = \frac{V}{-jX_C}$.

And later $\bar{I}$ can be computed using -

(i) $\bar{I} = \bar{I}_R + j\bar{I}_L$, (ii) $\bar{I} = \bar{I}_R - j\bar{I}_C$

(iii) $\bar{I} = \bar{I}_R + j(\bar{I}_C - \bar{I}_L)$



$$\bar{I}_Z = \frac{200}{10 + j15}$$

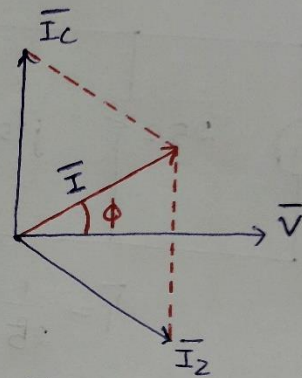
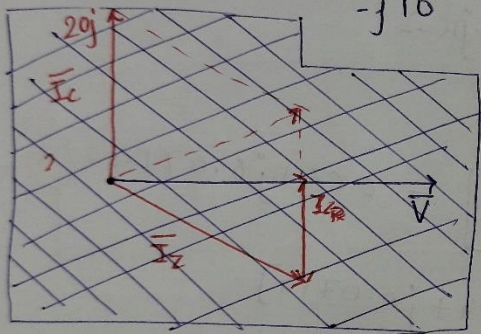
$$\bar{I}_Z = \frac{200}{\sqrt{10^2 + 15^2}} \angle \tan^{-1}\left(\frac{15}{10}\right)$$

$$\bar{I}_Z = 11.094 \angle -56.31^\circ \text{ A}$$

$$\bar{I}_Z = 11.094 (\cos 56.31 - j \sin 56.31)$$

$$= 6.154 - j9.2307$$

$$\bar{I}_C = \frac{200}{-j10} = 20j$$



• Phasor diagram •

$$\bar{I} = \bar{I}_Z + \bar{I}_C$$

$$\bar{I} = 6.154 - j9.2307 + j20$$

$$\bar{I} = 6.154 + j10.7693$$

$$\phi = \tan^{-1}\left(\frac{10.7693}{6.154}\right) = 60.254^\circ$$

$$\cos \phi = 0.496 \text{ (leading)}$$

* Quality factor :- Q :-

$$Q = 2\pi \times \frac{\text{maximum stored energy per cycle}}{\text{energy dissipated per cycle}}$$

$$\rightarrow \text{energy stored in capacitor} = \frac{1}{2} CV_c^2$$

$$\rightarrow \text{energy stored in inductor} = \frac{1}{2} LI_L^2$$

$$\text{Total energy stored} = \frac{1}{2} (CV_c^2 + LI_L^2).$$

In series R-L-C circuit, if current flowing through the ckt is $I = I_m \sin \omega t$ then, capacitor voltage $V_c =$

$$V_c = \frac{1}{C} \int I dt = \frac{1}{C} \int I_m \sin \omega t dt = -\frac{I_m \cos \omega t}{\omega C}$$

$$\therefore \text{Total energy} = \frac{1}{2} L I_m^2 \sin^2 \omega t + \frac{1}{2} C \frac{I_m^2 \cos^2 \omega t}{\omega^2 C^2}$$

$$\text{Resonant frequency } \omega_r = \frac{1}{\sqrt{LC}} \rightarrow \omega_r^2 = \frac{1}{LC}$$

$$\therefore \text{Total energy stored at resonance} = \frac{1}{2} L I_m^2 \sin^2 \omega t + \frac{1}{2} C \frac{I_m^2 \cos^2 \omega t}{\frac{1}{LC}}$$

$$= \frac{1}{2} L I_m^2 [\sin^2 \omega t + \cos^2 \omega t]$$

$$= \frac{1}{2} L I_m^2$$

I_m is the peak value of current $= \sqrt{2} I_{rms}$.

$$\therefore \text{Total energy stored at resonance} = \frac{L I_{rms}^2}{2}$$

$$\bullet \text{ Power dissipated} = I_{rms}^2 \times R.$$

$$\bullet \text{ Energy dissipated per cycle} = \frac{I_{rms}^2 \times R}{f_r} = \frac{R I_{rms}^2 \times 2\pi}{\omega_r}$$

$$Q = \frac{2\pi \times L I_{rms}^2}{\frac{R I_{rms}^2 \times 2\pi}{\omega_r}} = \frac{\omega_r L}{R}$$